

Input File Description

Program: projwfc.x / PWscf / Quantum ESPRESSO (version: 7.5)

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INTRODUCTION

Purpose of projwfc.x:

projects wavefunctions onto orthogonalized atomic wavefunctions, calculates Lowdin charges, spilling parameter, projected DOS (separated into up and down components for LSDA). Alternatively: computes the local DOS(E) integrated in volumes given in input (see [tdosinboxes](#)) or k-resolved DOS (see [kresolveddos](#)). Atomic projections are written to file "atomic_proj.xml".

Structure of the input data:

=====

&PROJWFC

...

/

Namelist: &PROJWFC

prefix CHARACTER

Default: 'pwscf'

prefix of input file produced by **pw.x** (wavefunctions are needed)

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outdir CHARACTER

Default: value of the ESPRESSO_TMPDIR environment variable if set; current directory ('.') otherwise

directory containing the input data, i.e. the same as in **pw.x**

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ngauss INTEGER

Default: 0

Type of gaussian broadening:

0 ... Simple Gaussian (default)

1 ... Methfessel-Paxton of order 1

-1 ... "cold smearing" (Marzari-Vanderbilt-DeVita-Payne)
 -99 ... Fermi-Dirac function

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degauss REAL

Default: 0.0

gaussian broadening, Ry (not eV!)

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Emin, Emax REAL

Default: (band extrema)

min & max energy (eV) for DOS plot

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DeltaE REAL

energy grid step (eV)

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lsym LOGICAL

Default: .true.

if **.true.** the projections are symmetrized,
 the partial density of states are computed
 if **.false.** the projections are not symmetrized, the partial
 DOS can be computed only in the k-resolved case

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diag_basis LOGICAL

Default: .false.

if **.false.** the projections of Kohn-Sham states are
 done on the orthogonalized atomic orbitals
 in the global XYZ coordinate frame.
 if **.true.** the projections of Kohn-Sham states are
 done on the orthogonalized atomic orbitals
 that are rotated to the basis in which the
 atomic occupation matrix is diagonal
 (i.e. local XYZ coordinate frame).

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pawproj LOGICAL

Default: .false.

if **.true.** use PAW projectors and all-electron PAW basis
 functions to calculate weight factors for the partial
 densities of states. Following Bloechl, [PRB 50, 17953 \(1994\)](#),
 Eq. (4 & 6), the weight factors thus approximate the real
 charge within the augmentation sphere of each atom.
 Only for PAW, not implemented in the noncolinear case.

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filpdos CHARACTER

Default: (value of [prefix](#) variable)

prefix for output files containing PDOS(E)

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filproj CHARACTER
Default: (standard output)
 file containing the projections

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lwrite_overlaps LOGICAL
Default: .false.
 if .true., the overlap matrix of the atomic orbitals prior to orthogonalization is written to "atomic_proj.xml".
 Does not work together with parallel diagonalization:
 for parallel runs, use "mpirun -np N projwfc.x -nd 1 ... "

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lbinary_data LOGICAL
Default: .false.
 CURRENTLY DISABLED.
 if .true., write atomic projections to a binary file.

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kresolvedDOS LOGICAL
Default: .false.
 if .true. the k-resolved DOS is computed: not summed over all k-points but written as a function of the k-point index.
 In this case all k-point weights are set to unity

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tdosinboxes LOGICAL
Default: .false.
 if .true. compute the local DOS integrated in volumes
 Volumes are defined as boxes with edges parallel to the unit cell, containing the points of the (charge density) FFT grid included within [irmin](#) and [irmax](#), in the three dimensions:
 from [irmin](#)(j,n) to [irmax](#)(j,n) for j=1,2,3 (n=1,[n_proj_boxes](#)).

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n_proj_boxes INTEGER
Default: 1
 number of boxes where the local DOS is computed

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irmin(i,n_proj_boxes), (i,n_proj_boxes) = (1,1) ... (3,n_proj_boxes) INTEGER
Default: 1 for each box
 first point of the given box
 BEWARE: [irmin](#) is a 2D array of the form: [irmin](#)(3,[n_proj_boxes](#))

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irmax(i,n_proj_boxes), (i,n_proj_boxes) = (1,1) ... (3,n_proj_boxes) INTEGER
Default: 0 for each box
 last point of the given box;
 (0 stands for the last point in the FFT grid)

BEWARE: `irmax` is a 2D array of the form: `irmax(3,n_proj_boxes)`

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plotboxes LOGICAL

Default: .false.

if `.true.`, the boxes are written in output as `xsf` files with 3D datagrids, valued 1.0 inside the box volume and 0 outside (visualize them as isosurfaces with isovalue 0.5)

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Notes

Format of output files

Projections are written to standard output, and also to file `filproj` if given as input.

The total DOS and the sum of projected DOS are written to file `"filpdos".pdos_tot`.

- * The format for the collinear, spin-unpolarized case and the non-collinear, spin-orbit case is:

```
E DOS(E) PDOS(E)
...
```

- * The format for the collinear, spin-polarized case is:

```
E DOSup(E) DOSdw(E) PDOSup(E) PDOSdw(E)
...
```

- * The format for the non-collinear, non spin-orbit case is:

```
E DOS(E) PDOSup(E) PDOSdw(E)
...
```

In the collinear case and the non-collinear, non spin-orbit case projected DOS are written to file `"filpdos".pdos_atm#N(X)_wfc#M(l)`, where N = atom number, X = atom symbol, M = wfc number, l=s,p,d,f (one file per atomic wavefunction found in the pseudopotential file)

- * The format for the collinear, spin-unpolarized case is:

```
E LDOS(E) PDOS_1(E) ... PDOS_2l+1(E)
...
```

where $LDOS = \sum_{m=1,2l+1} PDOS_m(E)$

and $PDOS_m(E)$ = projected DOS on atomic wfc with component m

- * The format for the collinear, spin-polarized case and the non-collinear, non spin-orbit case is as above with two components for both $LDOS(E)$ and $PDOS_m(E)$

In the non-collinear, spin-orbit case (i.e. if there is at least one fully relativistic pseudopotential) wavefunctions are projected onto eigenstates of the total angular-momentum.

Projected DOS are written to file `"filpdos".pdos_atm#N(X)_wfc#M(l_j)`, where N = atom number, X = atom symbol, M = wfc number, l=s,p,d,f and j is the value of the total angular momentum.

In this case the format is:

```
E LDOS(E) PDOS_1(E) ... PDOS_2j+1(E)
...
```

If `kresolveddos=.true.`, the k-point index is prepended to the formats above, e.g. (collinear, spin-unpolarized case)

```
ik E DOS(E) PDOS(E)
```

All $DOS(E)$ are in states/eV plotted vs E in eV

Orbital Order

Order of m-components for each l in the output:

1, cos(phi), sin(phi), cos(2*phi), sin(2*phi), .., cos(l*phi), sin(l*phi)

where phi is the azimuthal angle: $x=r \cos(\theta)\cos(\phi)$, $y=r \cos(\theta)\sin(\phi)$
This is determined in file upflib/ylmr2.f90 that calculates spherical harmonics.

for l=1:

```
1 pz      (m=0)
2 px      (real combination of m=+/-1 with cosine)
3 py      (real combination of m=+/-1 with sine)
```

for l=2:

```
1 dz2     (m=0)
2 dzx     (real combination of m=+/-1 with cosine)
3 dzy     (real combination of m=+/-1 with sine)
4 dx2-y2  (real combination of m=+/-2 with cosine)
5 dxy     (real combination of m=+/-2 with sine)
```

Defining boxes for the Local DOS(E)

Boxes are specified using the variables [irmin](#) and [irmax](#):

FFT grid points are included from $\text{irmin}(j,n)$ to $\text{irmax}(j,n)$
for $j=1,2,3$ and $n=1,\dots,n_{\text{proj_boxes}}$

[irmin](#) and [irmax](#) range from 1 to nr1 or nr2 or nr3

Values larger than $\text{nr1}/2/3$ or smaller than 1 are folded
to the unit cell.

If $\text{irmax} < \text{irmin}$ FFT grid points are included from 1 to irmax
and from irmin to $\text{nr1}/2/3$.

Important notices

The tetrahedron method is used if

- the input data file has been produced by pw.x using the option
occupations='tetrahedra', AND
- a value for degauss is not given as input to namelist &projwfc

* Gaussian broadening is used in all other cases:

- if [degauss](#) is set to some value in namelist &PROJWFC, that value
(and the optional value for ngauss) is used
- if [degauss](#) is NOT set to any value in namelist &PROJWFC, the
value of [degauss](#) and of [ngauss](#) are read from the input data
file (they will be the same used in the pw.x calculations)
- if [degauss](#) is NOT set to any value in namelist &PROJWFC, AND
there is no value of [degauss](#) and of [ngauss](#) in the input data
file, [degauss=DeltaE](#) (in Ry) and [ngauss](#)=0 will be used

Obsolete variables, ignored:

```
io_choice
smoothing
```